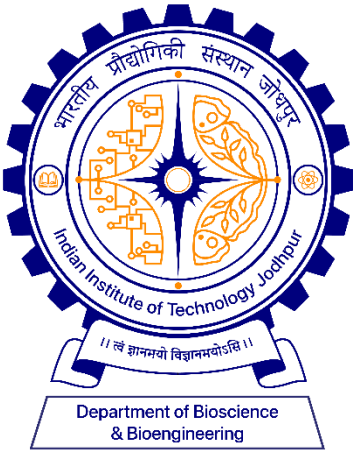
UG Curriculum



B. Tech in Bioengineering

1. Background

Bioengineering has emerged as one of the most transformative interdisciplinary domains of the 21st century, integrating principles of biology, engineering, computation, materials science, and data science to address pressing global challenges in healthcare, agriculture, energy, environment, and sustainability. Rapid advances in molecular biology, genomics, artificial intelligence, systems biology, biosensors, biomaterials, and precision medicine have significantly expanded the scope of bioengineering and necessitated a new generation of engineers equipped with multidisciplinary knowledge and translational skills. The proposed B.Tech. program in Bioengineering is designed to prepare students to understand, analyze, and engineer biological systems across multiple scales—from molecules and cells to tissues, organisms, and ecosystems. The curriculum combines strong foundations in biological sciences, engineering principles, computational methods, quantitative analysis, and hands-on laboratory training with emerging technologies such as machine learning, omics technologies, deep learning, bioinformatics, biosensors, and systems biology. The program emphasizes experiential learning, research-driven pedagogy, innovation, entrepreneurship, and industry-academia engagement. Students will gain exposure to real-world biological and engineering problems through laboratory modules, design courses, interdisciplinary electives, internships, and project-based learning. The curriculum is structured to foster critical thinking, creativity, ethical responsibility, and problem-solving abilities required for developing sustainable technological solutions. The Department of Bioscience and Bioengineering at Indian Institute of Technology Jodhpur aims to nurture highly skilled bioengineers capable of contributing to academia, research, healthcare industries, biotechnology enterprises, environmental sustainability, and national technological missions. The program also aligns with the vision of the National Education Policy (NEP 2020) by promoting multidisciplinary education, flexibility in learning pathways, innovation, and holistic student development.

2. Objectives of the Program

The B.Tech. program in Bioengineering aims to develop competent and socially responsible bioengineers equipped with strong interdisciplinary foundations, analytical capabilities, and innovation-driven mindsets to address complex biological and engineering challenges. The key objectives of the program are:

- To provide students with a strong foundation in biological sciences, engineering principles, mathematics, computation, and data-driven technologies relevant to modern bioengineering.
- To develop the ability to integrate engineering approaches with biological understanding for solving problems in healthcare, agriculture, energy, environment, and sustainability.
- To impart hands-on training in experimental techniques, computational tools, instrumentation, data analytics, and modern bioengineering methodologies.
- To equip students with skills in emerging areas such as bioinformatics, artificial intelligence, machine learning, systems biology, omics technologies, biomaterials, biosensors, and translational bioengineering.
- To foster critical thinking, innovation, creativity, and design-oriented problem-solving abilities through interdisciplinary learning and project-based education.
- To encourage research aptitude, scientific inquiry, and entrepreneurial thinking for the development of novel technologies and bio-based solutions.
- To develop effective communication, teamwork, leadership, and professional ethics necessary for multidisciplinary and multicultural work environments.
- To prepare graduates for higher studies, research careers, industrial leadership, start-up ventures, and societal contributions in the rapidly evolving bioeconomy.

3. Graduate Attributes

Graduates of the B.Tech. program in Bioengineering are expected to demonstrate the following attributes:

- Strong interdisciplinary knowledge in biology, engineering, computation, mathematics, and data sciences relevant to bioengineering applications.
- Ability to analyze complex biological systems and develop engineering-based solutions using experimental, computational, and analytical approaches.
- Proficiency in modern bioengineering tools and technologies including bioinformatics, machine learning, omics analysis, biosensors, biomaterials, and systems biology.
- Capability to design, develop, optimize, and implement innovative technologies for healthcare, diagnostics, therapeutics, agriculture, environmental sustainability, and industrial biotechnology.
- Competence in experimental design, data interpretation, quantitative reasoning, and evidence-based decision-making.
- Ability to work effectively as individuals and as members or leaders in multidisciplinary teams.

- Strong communication skills for effective scientific, technical, and societal engagement through written, oral, and digital platforms.
- Understanding of ethical, societal, environmental, and regulatory aspects associated with bioengineering research and applications.
- Entrepreneurial mindset and innovation-oriented thinking for translating scientific discoveries into scalable technologies and products.
- Commitment to lifelong learning, professional development, and adaptation to rapidly evolving scientific and technological advancements.

4. Learning Outcomes

Upon successful completion of the B.Tech. program in Bioengineering, graduates will be able to:

- Demonstrate comprehensive understanding of fundamental concepts in biology, bioengineering, mathematics, computing, and engineering sciences.
- Apply principles of molecular biology, biochemistry, microbiology, immunology, genetics, physiology, and systems biology to analyze biological processes and systems.
- Utilize computational and analytical tools including bioinformatics, artificial intelligence, machine learning, statistical methods, and data science for solving biological and biomedical problems.
- Design and conduct experiments, analyze experimental data, and interpret biological information using scientific and engineering methodologies.
- Develop engineering solutions related to diagnostics, therapeutics, biosensors, biomaterials, bioimaging, bioprocesses, and biomedical systems.
- Integrate multidisciplinary knowledge to address challenges in healthcare, precision medicine, environmental sustainability, agriculture, bioenergy, and industrial biotechnology.
- Apply engineering design principles to develop innovative, safe, sustainable, and socially relevant bioengineering technologies.
- Communicate technical concepts and research outcomes effectively through reports, presentations, scientific writing, and collaborative discussions.
- Demonstrate awareness of ethical responsibilities, biosafety practices, environmental sustainability, and societal impacts associated with bioengineering applications.
- Engage in independent learning, research, innovation, and entrepreneurial activities for continuous professional growth.

Table 1: Proposed B.Tech in Bioengineering Curriculum Structure

S. No.	Course Type	Course Category	Credits	Minimum Total
1	Institute Core (I)	Engineering (IE)	16	39 (27.6%)
		Science (IS)	17	
		HSS (IH)	06	
2	Programme Core (P)	Programme Compulsory (PC)	66	**90 (63.8%)**
		Programme Electives (PE)	18	
		(B.Tech.) Project (PP)	6	
3	Open (O)	Electives (OE)	12	12 (8.5%)
Total (Graded)				**141 (100%)**
4	Essential Audit (Non-taught)	Humanities (NH)	3	**12**
		Engineering (NE)	3	
		Industry-Academia Summer Internship outside IIT Jodhpur (two summers, minimum 45 days)	4	
		Design Credit	2	
Total for award of Degree (Graded + Non-Graded)				**153**

Institute-wide Engineering (IE) Courses (16 credits):

1. Engineering Mechanics [3-0-0]
2. Introduction to Computer Science [3-0-2]
3. Introduction to Machine Learning [3-0-2]
4. Engineering Workshop [0-0-2]
5. Engineering Drawing [0-0-2]
6. Environment and Sustainability [3-0-0]

Institute-wide Science (IS) courses (17 credits):

1. Physics [3-0-0]
2. Physics Lab [0-0-2]
3. Chemistry [3-0-0]
4. Chemistry Lab [0-0-2]
5. Mathematics I [2-1-0]
6. Mathematics II [2-1-0]
7. Introduction to Bioengineering [2-0-2]

Institute-wide NH courses (3 credits):

1. Communication Skill [0-0-2]
2. Introduction to Professional Ethics [1-0-0]
3. Performing Arts /Sports [0-0-1]
4. Social Connect and Responsibilities [0-0-1]

Institute-wide NE courses (3 credits):

1. Engineering Design I [0-0-2]
2. Engineering Design II [0-0-2]
3. Introduction to Profession I [0-0-1]
4. Introduction to Profession II [0-0-2]

Table 2. Semester-wise plan for B.Tech in Bioengineering

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
	I Semester (ME/BB)/ II Semester (CS/EE)						**II Semester (ME/BB)/ I Semester (CS/EE)**				
IE	Engineering Mechanics	3-0-0	3	-	3	IE	Introduction to Machine Learning	3-0-2	5	-	4
IE	Introduction to Computer Science	3-0-2	5	-	4	IS	Chemistry	3-0-0	3	-	3
IS	Physics	3-0-0	3	-	3	IS	Introduction to Bioengineering	2-0-2	4	-	3
IH	HSS I	3-0-0	3		3	IS	Chemistry Lab	0-0-2	2	-	1
IS	Physics Lab	0-0-2	2	-	1		Mathematics II	2-1-0	3	-	3
IS	Mathematics I	2-1-0	3	-	3	IS	Engineering Workshop	0-0-2	1	-	1
IE	Engineering Drawing	0-0-2	2	-	1	IE	Engineering Design II	0-0-2	2	1	-
NE	Engineering Design I	0-0-2	2	1	-	NE	HSS II	3-0-0	3		3
NH	Communication Skill	0-0-2	2	1	-	IH	Introduction to Profession	0-0-2	2	1	-
NH	Social Connect and responsibilities	0-0-1	1	0.5	-	NE					
NH	Performing Arts /Sports	0-0-1	1	0.5	-						
	Total				**18**		**Total**				**18**
	III Semester						**IV Semester**				
PC	Probability, statistics and stochastic process	3-1-0	4	-	4	PC	Biophysics and Structural Biology	3-0-0	3	-	3
PC	Data Structures and Algorithms	3-0-0	3	-	3	PC	Biology of Cells	3-0-0	3	-	3
PC	Biochemistry	3-0-0	3	-	3	PC	Signals and Systems	3-1-0	4	-	4
PC	Bioinformatics	3-0-0	3	-	3	PC	Analytical Techniques for Bioengineering	3-1-0	4	-	4
PC	Bioengineering Lab-I	0-0-6	6	-	3	PC	Quantitative Physiology	3-0-0	3	-	3
IE	Environment and Sustainability	3-0-0	3	-	3	PC	Bioengineering Lab-II	0-0-6	6	-	3
	Total				**19**		**Total**			**-**	**20**
	Summer Semester Internship (Minimum 45 days; Outside IIT Jodhpur)										
	V Semester						**VI Semester**				
PC	Molecular Biology & Genetic Engineering	3-0-0	3	-	3	PC	Introductory Omics	3-0-0	3	-	3
PC	Deep Learning	3-0-0	3	-	3	PC	Biosensors and Instrumentation	3-0-0	3	-	3
PC	Translational Immunology	3-0-0	3	-	3	PC	Bioprocess Engineering	3-0-0	3	-	3

PC	Biomaterials Engineering	3-0-0	3	-	3	PC	Systems Biology	3-0-0	3	-	3
PC	Bioengineering Lab-III	0-0-6	6	-	3	PC	Bioengineering Lab-IV	0-0-6	6	-	3
PE	Programme Elective	3-0-0	3	-	3	PE	Program Elective	3-0-0	3	-	3
NH	Professional Ethics I	1-0-0	1	1	-						
Total		**15-3-8**	**26**	**1**	**18**	**Total**		**15-3-8**	**26**	**1**	**18**
Summer Semester Internship (Minimum 45 days; Outside IIT Jodhpur)											
	VII Semester						**VIII Semester**				
PP	B.Tech Project 1	0-0-3	-		3	PP	B.Tech Project 2	0-0-3	-		3
PE	Programme Elective	3-0-0	3	-	3	PE	Programme Elective	3-0-0	3	-	3
PE	Programme Elective	3-0-0	3	-	3	PE	Programme Elective	3-0-0	3	-	3
OE	Open Elective	3-0-0	3	-	3	OE	Open Elective	3-0-0	3	-	3
OE	Open Elective	3-0-0	3	-	3	OE	Open Elective	3-0-0	3	-	3
				-							
Total					**15**	**Total**				**-**	**15**
Total Graded and Non-Graded Credit										**6**	**141**
Non-Graded Design Credits										**4**	**-**
Grand Total										**12**	**153**

Table 3: Curriculum Structure B.Tech with Minor in Bioengineering

S. No.	Course Type	Course Category	B.Tech. Regular	Dual Degree (B.Tech +PhD)	B.Tech. With Minor
1	Institute Core (I)	Engineering (IE)	16	16	16
		Science (IS)	17	17	17
		Humanities (IH)	06	06	06
3	Programme Core (P)	Programme Compulsory (PC)	66	66	66
		Programme Electives (PE)	18	18	18
		B.Tech. Project (PP)	6	6	6
4	Open (O)	Electives (OE)	12	12	0
5	Additional Credits	Programme Compulsory (PC)	0		18
6	PhD	Compulsory (DC)		12	0
		Electives (DE)		12	
		PhD. Thesis (MP)		64	
Total Graded			**141**	**229**	**147**
7	Essential Audit (Non-taught)	Humanities (NH)	3	3	3
		Engineering (NE)	3	3	3
		Design/Practical Experience (ND)	2	2	2
		Industry-Academia Summer Internship outside IIT Jodhpur (two summers, minimum 45 days)	4	4	4
Total Graded + Essential Audit			**153**	**241**	**159**

Table 4: Semester-wise Structure: B.Tech Programme with Minor in Bioengineering

Cat	*Course*	*LTP*	*CH*	*NC*	*GC*	*Cat*	*Course*	*LTP*	*CH*	*NC*	*GC*
I-IV Semester (Same as Table 4):											
	V Semester						**VI Semester**				
PC	**Programme Compulsory**			- - -	16	PC	**Programme Compulsory**			- - -	16
PE/ OE	Programme/Open Elective/Minor Compulsory courses	3-0-0	3	-	3	PP					
NH	Professional Ethics I	1-0-0	1	1	-	PE/SC	Program elective/Minor compulsory courses	3-0-0			3
	Total	**15-3-8**	**26**	**1**	**19**		**Total**	**15-3-8**	**26**	**1**	**19**
	VII Semester						**VIII Semester**				
	VII Semester						**VIII Semester**				
PP	B.Tech Project	0-0-3			3	PP	B.Tech Project	0-0-3			3
PE/ OE	Programme/Open Electives/Minor Compulsory Courses	15-0-0	15	-	15	PE/ OE	Programme/Open Electives/Minor Compulsory Courses	15-0-0		-	15
	Total			**-**	**18**		**Total**			**-**	**18**
											147
										Essential Audit	**12**
										Grand Total	**159**

1. List of Programme Compulsory (PC) Courses for B.Tech in Bioengineering

Table 5: Focus area wise Program Compulsory (PC) courses

S. No.	Focus Area	PC Course	L-T-P	Credit
1	Foundations for Bioengineering	Biology of Cells	3-0-0	3
		Biochemistry	3-0-0	3
		Biophysics and Structural Biology	3-0-0	3
		Probability, Statistics and Stochastic Processes	3-1-0	4
2	Engineering Biological Processes and Systems	Introductory Omics	3-0-0	3
		Molecular Biology and Genetic Engineering	3-0-0	3
		Bioprocess Engineering	3-0-0	3
		Biomaterials Engineering	3-0-0	3
		Translational Immunology	3-0-0	3
3	Analytical Bioengineering	Signals and Systems	3-0-0	3
		Quantitative Physiology	3-0-0	3
		Biosensors and Instrumentation	3-0-0	3
		Analytical Techniques for Bioengineering	3-1-0	4
4	Computational Bioengineering	Data Structures and Algorithms	3-0-0	3
		Bioinformatics	3-1-0	4
		Deep Learning	3-0-0	3
		Systems Biology	3-0-0	3

2. List of Programme Elective (PE) Courses for B.Tech in Bioengineering

Table 6: Focus area wise Program Elective (PE) courses

Sr. No.	PE Course	Level	L-T-P	Credit
1	Tissue Engineering	400	3-0-0	3
2	Modern Approaches of Drug Designing	400	3-0-0	3
3	Microbial Remediation and Environmental Bioengineering	400	3-0-0	3
4	Synthetic Biology	700	3-0-0	3
5	Plant Genetic Engineering and Functional Genomics	700	3-0-0	3
6	Bioenergy Systems	700	3-0-0	3
7	Viral Infections and Antiviral Drug Development	400	3-0-0	3
8	Introduction to Chemical Biology	700	3-0-0	3
9	Neuroengineering Systems	700	3-0-0	3
10	Digital Image Processing	400	3-0-0	3
11	Bioimaging Applications	700	3-0-0	3
12	Biomedical Signals and Devices	400	3-0-0	3
13	Theranostic Systems	700	3-0-0	3
14	BigData and AI Applications in Bioengineering	700	3-0-0	3
15	Computational Methods for Multi-Omics	400	3-0-0	3
16	Precision P4 Medicine	700	3-0-0	3
17	Computational Structural Bioinformatics	700	3-0-0	3
18	Computational Genomics	700	3-0-0	3

Note: 400 level courses are specific to B.Tech. students, while 700 level courses are open to B.Tech., M.Tech. and Ph.D. students.

3. List of Programme Compulsory (PC) Courses for Minor in Bioengineering

Table 7: Focus area wise Program Elective (PC) courses for Minor in Bioengineering

Sr. No.	PC Course	L-T-P	Credit
1	Biology of Cells	3-0-0	3
2	Biochemistry	3-0-0	3
3	Biomaterials Engineering	3-0-0	3
4	Quantitative Physiology	3-0-0	3
5	Biosensors and Instrumentation	3-0-0	3
6	Bioinformatics	3-1-0	4